Weather, Soil & Water Modeling

TEAM DEFINITIONS & HANDOFF GLOSSARY

Smart Agricultural Water Management System — Rice / Nile Delta

Purpose
A practical reference for teammates, presenters, developers, and anyone using the weather–soil–crop water outputs. It explains the important terms, units, file fields, formulas, status labels, and what must NOT be presented as a field measurement.

Fastest rule to remember

ET_0 → reference atmosphere demand | ETc → rice crop demand | Irrigation requirement → water that must be supplied after rainfall and rice-specific losses

Where everything lives — and what counts as the model

Term / field	Plain-English definition	How your team should use it
The “model”	There is no single .pkl/.h5 AI model file. The project model is a scientific Python pipeline: cleaned inputs + crop parameters + FAO-56 ET_0 + ETc + irrigation-water balance + validation.	For software integration, use the reusable functions in src/. For full reproducibility and explanation, read/run the notebooks in order 01→06.
src/	Source code: clean reusable Python functions extracted from the notebooks.	Website/chatbot/optimization teammates should call these functions instead of copying notebook cells.
src/weather_preprocessing.py	Reusable weather-cleaning/preparation helpers.	Use when reproducing weather preprocessing programmatically.
src/soil_preprocessing.py	Reusable soil/AWC preparation helpers.	Use when reproducing soil preprocessing programmatically.
src/crop_parameters.py	Reusable rice growth-stage/Kc logic.	Use when assigning stages or crop coefficients in another system.
src/evapotranspiration.py	Reusable evapotranspiration calculation code, including FAO-56 ET_0 logic.	This is a core model-code file for atmospheric/crop water-demand calculations.
src/irrigation_requirement.py	Reusable net/gross irrigation-requirement calculations and water-balance logic.	This is the main model-code file for irrigation recommendation logic.
src/water_volume.py	Reusable conversion from irrigation depth (mm) and area (feddans) to volume (m^3).	Use in the web app/chatbot when the user supplies farm area.
data/processed/	Cleaned, modeling-ready, and calculated output datasets.	This is the main folder teammates should use for analysis/integration. It contains cleaned weather/soil and final ET_0, ETc, irrigation, and validation outputs.
data/reference/	Scientific/reference tables used by the calculations: crop calendar, Kc/stage parameters, seasons, parameter sources, and varieties.	Use when reproducing or explaining assumptions. These are reference inputs, not final calculated results.
notebooks/	The six step-by-step Jupyter/Colab notebooks that show how the pipeline was built, checked, and validated.	Use for reproducibility, auditing, learning, or rerunning the workflow. They are not the preferred production interface for the web app.
tests/	Automated tests for ET_0, irrigation requirement, and water-volume functions.	Run tests when code changes to make sure core calculations still behave correctly.
reports/	Human-readable documentation: data-quality report, methodology, validation report, and final documentation.	Read before presenting results or using irrigation estimates as evidence.
README.md	Short guide to the component, structure, outputs, and how to use it.	Start here if a teammate is new to the folder.
AgriWater/AgriWater_Clean_Data/data/table_exports/	Shared project source/export tables used as raw inputs by this component.	Do not duplicate the large raw data inside weather_soil_water_modeling/. The notebooks read from this shared location.

Fast folder rule

processed = cleaned/calculated outputs to USE | reference = scientific parameters/assumptions | notebooks = how the work was done | src = reusable production code | reports = what the results mean

1. Core concepts everyone should know

Term / field	Plain-English definition	How your team should use it
Water requirement estimation pipeline	The full sequence that turns weather, soil, and rice-growth information into estimated irrigation-water needs.	Use this phrase for the whole component. It is a scientific estimation pipeline, not a sensor measuring farm water use.
Scenario estimate	A calculated result produced under documented assumptions.	Use for irrigation results. Do NOT call it “actual irrigation used” unless field measurements exist.
Field measurement	A value directly measured at a farm or experiment using instruments, meters, observations, or a documented experimental protocol.	Treat as stronger evidence than a theoretical scenario, but still check unit, study conditions, and quality.
Gridded estimate	A value estimated for a geographic grid cell rather than measured at one exact farm.	NASA POWER weather and mapped soil values are spatial estimates. Do not claim they came from a farmer’s field sensor.
Location / grid point	One spatial point represented by point_id with latitude and longitude.	Use point_id as the main geographic join key across weather, ET_0, ETc, and water outputs.
Season	The documented rice growing period for a specific year.	The rice ETc/irrigation calculations use documented seasons 2017–2023; do not silently extend the crop calendar to later years.
Sensitivity analysis	Recalculating results under different plausible assumptions to see how much outputs change.	Used for seepage/percolation and irrigation-efficiency scenarios. It shows uncertainty; it is not a second set of measurements.
Validation	Checking whether calculated results are reasonable by comparing them with suitable independent references or measurements.	A comparison can reveal disagreement. Validation does not mean the model automatically “passes.”
Consistency check	A comparison against another model-generated scenario or calculation.	Useful for debugging/understanding differences, but weaker than independent field/literature validation.
Calibration	Adjusting model parameters using observations so predictions better match evidence.	Not the same as validation. The current pipeline did not claim field calibration.

Critical wording for presentations

Say: “estimated seasonal irrigation requirement” or “scenario-based water requirement.” Avoid: “the farm used X m³” or “actual farmer consumption” because the project does not have field measurements proving that.

2. Weather data definitions

Term / field	Plain-English definition	How your team should use it
NASA POWER	NASA POWER provides meteorological and solar-energy data, often as gridded estimates.	It is the weather source used by this component. Treat it as spatial weather data, not farm-station measurements.
T2M	Mean air temperature at 2 metres above the surface. Unit: °C.	Used as a main FAO-56 ET_0 temperature input.
T2M_MAX	Daily maximum air temperature at 2 m. Unit: °C.	Used in saturation vapour pressure and radiation calculations.
T2M_MIN	Daily minimum air temperature at 2 m. Unit: °C.	Used with T2M_MAX in ET_0 calculations.
RH2M	Relative humidity at 2 m. Unit: %.	Used to estimate actual vapour pressure because RHmax/RHmin were not available.

Term / field	Plain-English definition	How your team should use it
PRECTOTCORR	Corrected daily precipitation / rainfall. Unit: mm/day.	Used to calculate seasonal rainfall and effective rainfall.
WS2M	Wind speed at 2 m. Unit: m/s.	Direct FAO-56 Penman–Monteith input.
PS	Surface atmospheric pressure. Unit: kPa.	Used to calculate the psychrometric constant in ET_0.
ALLSKY_SFC_SW_DWN	All-sky downward shortwave solar radiation at the surface. Unit: MJ/m²/day in the project dataset.	Used as incoming solar radiation (Rs) in ET_0.
obs_date	Date associated with a daily weather record.	Use with point_id to join daily weather to ET_0/ETc.
rainfall_review_flag	A quality-review flag for unusually high rainfall values; flagged rows were kept rather than deleted automatically.	A flag means “review this value,” not “this value is wrong.”
et0_inputs_complete	Boolean showing whether all required ET_0 weather inputs are available for a row.	Useful for filtering calculation-ready weather records.
rainfall_available	Boolean showing whether rainfall exists for that row.	Used during cleaning to retain rows suitable for irrigation calculations.

3. Soil and available-water definitions

Term / field	Plain-English definition	How your team should use it
Soil texture	The relative proportions of sand, silt, and clay.	Used to describe drainage and water-retention behaviour; do not infer exact field hydraulic behaviour from texture alone.
sand_pct	Sand fraction. Unit: % by weight.	Higher sand generally indicates faster drainage / lower water holding, but use project data and documented assumptions rather than stereotypes.
silt_pct	Silt fraction. Unit: % by weight.	Part of soil texture; combined with sand and clay should total about 100% for valid mapped soil records.
clay_pct	Clay fraction. Unit: % by weight.	Used in the soil-adjusted seepage sensitivity scenario.
bulk	Bulk density. Unit: g/cm³.	A soil physical property related to compaction and porosity. It is not itself soil moisture.
organic_carbon	Organic carbon content. Unit: % by weight.	Soil property that can influence structure and water behaviour; it is mapped/estimated, not laboratory field data in this project.
ph_water	Soil pH measured/represented in water; unitless.	Describes acidity/alkalinity. Do not treat pH as salinity or EC.
AWC	Available Water Capacity: water that soil can store and make available to plants between relevant moisture limits.	Capacity is not the same as water currently stored in the soil on a specific date.
awc_smu_mm_per_m	Mapped AWC for the soil mapping unit. Unit: mm of plant-available water per metre of soil.	Use as a soil capacity property; do not subtract it from irrigation need as if all of it were present at planting.
awc_rootable_mm_estimate	Estimated available water over the rootable soil depth. Unit: mm.	A capacity estimate. Actual stored soil water requires initial moisture information, which was unavailable.
SMU / smu_id	Soil Mapping Unit identifier.	Links mapped soil properties belonging to the same mapped unit.

Term / field	Plain-English definition	How your team should use it
exclude_from_model	Flag showing a soil record should not be used in soil-based modeling.	Two points were excluded because they represented non-soil/water-body mapping issues.
PROVISIONAL_NOT_FIELD_VALIDATED	Status indicating a soil/AWC value is usable as an estimate but has not been validated by local field sampling.	Always communicate this limitation when soil values influence recommendations.

Do not confuse AWC with stored soil water

AWC describes how much water the soil could hold for plants. It does NOT tell us how wet the soil actually was at transplanting. Therefore the final irrigation model did not deduct AWC as actual initial stored water.

4. Rice crop and growth-stage definitions

Term / field	Plain-English definition	How your team should use it
Kc (crop coefficient)	A dimensionless factor that converts reference ET_0 into crop ETc: $ETc = ET_0 \times Kc$.	Use stage-appropriate Kc values; do not use one generic value for the whole season unless explicitly documented.
Initial stage	Early crop stage. Project reference Kc ≈ 1.05; reference stage length 30 days.	Used for days 1–30 of the modeled 150-day rice season.
Development stage	Transition stage when Kc rises from initial toward mid-season. Reference stage length 30 days.	Kc was interpolated linearly from 1.05 to 1.20 rather than inventing a fixed development Kc.
Mid-season stage	Main active crop stage. Project reference Kc ≈ 1.20; reference stage length 60 days.	Used for days 61–120.
Late-season stage	Final crop stage; Kc decreases. Reference endpoint range 0.60–0.90; stage length 30 days.	Low/high Kc scenarios preserve source uncertainty instead of forcing one endpoint.
growth_stage	Label for the rice stage assigned to a daily record.	Important join/analysis field in ETc outputs.
season_length_days	Expected duration of a rice variety/season where available.	Some varieties have missing duration; missing values were not fabricated.
variety	A named rice cultivar.	Crop-water parameters in the source table were general rice parameters, not variety-specific unless explicitly stated.
SAT / soil saturation requirement	Water needed to saturate the soil during paddy preparation. Project reference: 200 mm.	Included as a rice-specific component of net seasonal irrigation requirement.
WL / water-layer establishment	Water used to establish the standing water layer in paddy rice. Project reference: 100 mm.	Included in the rice-specific water-balance scenario.
PERC / seepage-percolation rate	Water lost downward/sideways from a flooded rice field. Project base range: 4–8 mm/day.	Major uncertainty driver. A separate clay-adjusted sensitivity scenario was also calculated.

5. Evapotranspiration and FAO-56 terms

Term / field	Plain-English definition	How your team should use it
ET_0 (reference evapotranspiration)	Atmospheric evaporative demand from a standardized reference surface. Unit: mm/day.	It is climate demand, not rice water requirement by itself.

Term / field	Plain-English definition	How your team should use it
FAO-56 Penman–Monteith	Standard equation used to calculate ET_0 from radiation, temperature, humidity, wind, and pressure.	This is the ET_0 method implemented in Notebook 04 and reusable code.
ETc (crop evapotranspiration)	Estimated water consumed by the rice crop through evaporation + transpiration: ETc = ET_0 × Kc. Unit: mm/day or mm/season after aggregation.	ETc is crop consumption, not total irrigation supply. Rice irrigation also includes water losses/management needs.
es	Saturation vapour pressure: maximum water-vapour pressure air can support at a given temperature. Unit: kPa.	Intermediate FAO-56 variable.
ea	Actual vapour pressure: actual water-vapour pressure in the air. Unit: kPa.	Estimated from mean relative humidity in this project.
VPD	Vapour Pressure Deficit = es − ea. Unit: kPa.	Represents atmospheric drying demand; larger VPD generally increases evapotranspiration demand.
Δ (delta)	Slope of the saturation vapour pressure curve. Unit: kPa/°C.	Intermediate FAO-56 variable based on temperature.
γ (gamma)	Psychrometric constant. Unit: kPa/°C.	Calculated from surface atmospheric pressure.
Ra	Extraterrestrial radiation at the top of the atmosphere. Unit: MJ/m²/day.	Calculated from latitude and day of year.
Rs	Incoming solar radiation at the surface. Unit: MJ/m²/day.	Taken from ALLSKY_SFC_SW_DWN.
Rso	Clear-sky solar radiation. Unit: MJ/m²/day.	Used in net longwave-radiation calculation; near-sea-level approximation was used.
Rns	Net shortwave radiation. Unit: MJ/m²/day.	Incoming shortwave radiation after accounting for reference-surface albedo.
Rnl	Net outgoing longwave radiation. Unit: MJ/m²/day.	Thermal radiation loss from the surface/atmosphere system.
Rn	Net radiation = Rns − Rnl. Unit: MJ/m²/day.	Energy available for evapotranspiration in FAO-56.
G	Soil heat flux. Unit: MJ/m²/day.	For daily ET_0 calculations, G was taken as 0 following the daily FAO-56 convention.

6. Irrigation-water definitions and formulas

Term / field	Plain-English definition	How your team should use it
Effective rainfall (Pe)	The part of rainfall that actually contributes to crop/field water need after unavoidable losses. Unit: mm.	The model used a documented FAO-style monthly effective-rainfall method, then allocated it back to rainy days.
Seasonal rainfall	Total precipitation during the modeled rice season. Unit: mm/season.	Not the same as effective rainfall; only the effective fraction reduces irrigation need.
Net irrigation requirement	Estimated water needed at the field after effective rainfall, including rice-specific requirements/losses. Unit: mm/season.	Main field-level requirement. Base formula: ETc + SAT + PERC + WL − Pe.
Gross irrigation requirement	Water that must be supplied by the irrigation system before accounting for delivery/application losses.	Gross = Net / irrigation efficiency. Because local measured efficiency was unavailable, 60%, 70%, and 80% were sensitivity scenarios.
Irrigation efficiency	Fraction of supplied water that effectively reaches/serves the field requirement.	An assumption/sensitivity parameter here, not a locally measured farm value.
Stored soil water adjustment	Potential reduction in irrigation need if reliable initial soil moisture is known.	Set to 0 in the final model because initial field moisture measurements were unavailable.

Term / field	Plain-English definition	How your team should use it
Seepage	Lateral water movement/leakage out of a flooded field or bund system.	Combined with percolation in the project water-loss parameterization.
Percolation	Downward water movement through the soil below the effective field/root zone.	A major paddy-rice water-loss term and major uncertainty source.
Drainage	Water intentionally or naturally leaving the field via surface/subsurface pathways.	Should only be included if a documented dataset/method supports it; do not invent a drainage loss.
Water depth	Equivalent vertical depth of water over an area. Unit: mm.	Useful for agronomic calculations and converting to volume.
Water volume	Physical amount of water. Unit: m^3.	Project conversion: Volume (m^3) = depth (mm) × area (feddans) × 4.2.
Feddan	Egyptian land-area unit used in the project. 1 feddan ≈ 4,200 m^2.	This is why 1 mm over 1 feddan equals about 4.2 m^3.
mm/day	Daily water depth rate.	Used for ET_0, ETc, rainfall, and seepage/percolation rates.
mm/season	Accumulated water depth over a rice season.	Used for seasonal ETc and irrigation requirements.
m^3/feddan	Water volume normalized per feddan.	Best handoff unit for farm/application-facing outputs when area is known or standardized.
Low / high scenario	Lower and upper modeled requirements produced from uncertain Kc/seepage assumptions.	Keep the range. Do not silently replace it with one “exact” number.

Main rice irrigation equation

Net irrigation requirement = ETc + soil saturation requirement + seepage/percolation + water-layer establishment − effective rainfall. Gross requirement is calculated separately by dividing the net requirement by an assumed irrigation efficiency.

7. Validation and status labels

Term / field	Plain-English definition	How your team should use it
NOT_FIELD_MEASURED	The output is a modeled estimate, not a direct farm measurement.	Keep this status in downstream apps/analysis so users understand provenance.
FAO_BASED_SCENARIO_NOT_FIELD_VALIDATED	The main irrigation requirement uses FAO-based methods/parameters but was not validated against local field measurements.	Appropriate label for the main model output.
PROJECT_SOIL_SENSITIVITY_NOT_FIELD_VALIDATED	A secondary soil/clay-adjusted scenario used to test sensitivity.	Do not present it as the “true” seepage value.
SENSITIVITY_ANALYSIS_NOT_MEASURED	Gross irrigation outputs vary assumed irrigation efficiency rather than using a measured efficiency.	Display assumptions if these values are shown to farmers or decision-makers.
SCENARIO_ESTIMATE_NOT_FIELD_VALIDATED	Status found in existing project scenario tables.	Do not use those scenario values as ground-truth ML labels.
BENCHMARK_DISCREPANCY_REQUIRES_CAUTION	The modeled irrigation range did not overlap the chosen literature benchmark range.	This is an important warning: report the discrepancy and avoid claiming field accuracy.
Literature benchmark	Published/experimental values converted to compatible irrigation units for comparison.	Use for context/validation, but verify study method, variety, location, and irrigation treatment before one-to-one comparisons.
0% literature overlap	None of the modeled seasonal low–high ranges overlapped the 5th–95th percentile literature benchmark used in the notebook.	This does not mean code failure. It means the modeled scenario is higher and requires cautious interpretation/local validation.

8. Join keys and fields teammates will use

Term / field	Plain-English definition	How your team should use it
point_id	Stable identifier for each geographic grid point.	Primary spatial join key across weather, ET_0, ETc, soil, and irrigation outputs.
latitude / longitude	Geographic coordinates of a grid point.	Use for mapping and linking satellite pixels/areas to model locations.
obs_date	Daily date.	Primary time join key for weather, ET_0, and daily ETc.
season_year	Year of the documented rice season.	Use for seasonal ETc and irrigation summaries.
day_in_season	Day number within the modeled rice season.	Used to assign growth stage and interpolate Kc.
growth_stage	Initial, development, mid-season, or late-season.	Useful for satellite vegetation-stage comparisons and irrigation scheduling logic.
ET0_mm_day	Daily reference evapotranspiration.	Use as atmospheric demand feature.
ETc_low_mm_day / ETc_high_mm_day	Daily rice crop evapotranspiration low/high estimates.	Use for crop-water analysis; keep uncertainty range where relevant.
net_irrigation_low_mm / net_irrigation_high_mm	Seasonal field-level irrigation requirement range.	Main decision-support water-depth outputs.
net_low_m3_per_feddan / net_high_m3_per_feddan	Seasonal field-level irrigation volume range per feddan.	Main user-facing volume outputs for farm-area scaling.

9. Which output file should each teammate use?

Term / field	Plain-English definition	How your team should use it
weather_daily_clean.csv.gz	Clean daily weather with grid/location fields and quality flags.	Satellite + ML/DL: join by point_id + obs_date. Use as weather features.
soil_clean.csv	Usable cleaned soil/AWC records (36 points).	ML/DL + optimization: static soil features by point_id. Remember two grid points have no usable soil record.
rice_growth_stage_kc_reference.csv	Reference stage lengths and Kc ranges.	Anyone reproducing ETc or explaining crop-stage logic.
eto_daily.csv.gz	Daily FAO-56 ET_0 outputs plus intermediate variables.	Satellite + ML/DL + modeling: atmospheric evaporative demand by point/date.
etc_daily.csv.gz	Daily rice ETc with growth stage and Kc low/high.	Satellite + optimization: crop-water demand by point/date/stage.
etc_seasonal.csv	Seasonal ETc totals for each point and season.	Dashboards, summaries, optimization, reporting.
irrigation_requirement_seasonal.csv	Final seasonal irrigation requirement calculations, sensitivity fields, volumes, and statuses.	Optimization + chatbot + web app: MAIN water requirement output. Preserve status/assumption fields.
irrigation_validation_summary.csv	Compact validation findings and limitations.	All teammates: read before presenting or training on irrigation outputs.
literature_irrigation_benchmark.csv	Cleaned/converted literature irrigation records used for benchmarking.	Validation/research only; do not treat as automatically comparable ground truth.

10. Team-specific handoff cheat sheet

Satellite / image-processing teammate

- Join using point_id and obs_date (plus latitude/longitude when spatial matching is needed).
- Use weather_daily_clean.csv.gz, eto_daily.csv.gz, and etc_daily.csv.gz.
- Compare NDVI/EVI with growth_stage, ET_0, ETc, rainfall, and temperature.
- Remember all weather values are gridded estimates, not farm sensors.

Data cleaning / ML-DL teammate

- Use cleaned weather and soil features with their documented units and quality/status fields.
- Do NOT use existing water_scenario values or the new irrigation scenario estimates as unquestioned ground-truth labels.
- If modeling water demand, keep scenario provenance and validation status as metadata.
- There are 38 weather grid points but only 36 usable soil points; handle the two missing-soil locations explicitly.

Optimization / chatbot / web-app teammate

- Main file: irrigation_requirement_seasonal.csv.
- Prefer displaying a low–high range rather than one exact number.
- For farm area A feddans: volume = irrigation depth (mm) × A × 4.2.
- Clearly label results as estimates/scenarios and expose the validation/uncertainty warning.
- Do not say the value is actual farmer consumption or a field measurement.

Presentation / report teammate

- ET_0 = climate/reference demand; ETc = rice crop demand; irrigation requirement = ETc plus rice-specific water needs/losses minus effective rainfall.
- The final model is reproducible and scientifically structured, but not locally field-validated.
- A key validation finding is the model's irrigation range was higher than the selected literature benchmark, so conclusions must be cautious.

11. "Do not confuse these" quick table

Term / field	Plain-English definition	How your team should use it
ET_0 vs ETc	ET_0 = reference atmospheric demand. ETc = rice crop demand after applying Kc.	ETc = ET_0 × Kc.
ETc vs irrigation requirement	ETc only represents crop evapotranspiration. Irrigation requirement also accounts for rainfall, paddy preparation, standing water, and seepage/percolation.	Never use ETc alone as "how much irrigation water to supply."
Net vs gross irrigation	Net = field requirement. Gross = source/system supply before efficiency losses.	Gross is larger when efficiency < 100%.
Rainfall vs effective rainfall	Rainfall = total precipitation. Effective rainfall = usable part that reduces irrigation need.	Do not subtract all rainfall automatically.
AWC vs actual stored water	AWC = storage capacity. Actual stored water = current moisture available at a specific time.	The project did not have enough initial-moisture data to deduct stored soil water.
Scenario vs measurement	Scenario = calculated under assumptions. Measurement = observed directly.	Always preserve status labels.
Validation vs agreement	Validation means testing against evidence. A disagreement is still a valid validation finding.	The 0% overlap is a result to report, not hide.

12. One-minute quick reference

Three formulas teammates should know

1) ETc = ET_0 × Kc
2) Net irrigation ≈ ETc + SAT + seepage/percolation + WL − effective rainfall
3) Water volume (m^3) = water depth (mm) × area (feddans) × 4.2

Item	Project reference
Main spatial key	point_id
Main daily key	point_id + obs_date
Main seasonal key	point_id + season_year
Weather coverage	38 grid points; daily weather 2017–2026 (latest period may be incomplete)
Rice modeled seasons	2017–2023 documented seasons
Usable soil coverage	36 of 38 grid points
Main irrigation output	irrigation_requirement_seasonal.csv
Main warning	Scenario-based estimate; not actual field consumption; literature benchmark discrepancy requires caution

Final handoff statement: The component estimates rice water requirements using cleaned gridded weather, mapped soil properties, documented crop stages/Kc values, FAO-56 ET_0, ETc, effective rainfall, and rice-specific water-loss/management scenarios. Outputs are reproducible and integration-ready, but irrigation estimates must remain clearly labeled as scenarios until suitable local field measurements are available for validation.